

Multiple Choice (10 marks)

1. Find the Laplace transform $L\{g(t)\}$, where $g(t) = 0, t < 4$ and $g(t) = (t - 4)^2, t \geq 4$

$(2e^{-4s}) / s^3$

(a) $2 / s^3$

(b) $(2 / s^2) * e^{-4s}$

(c) $4 / s^3$

$(2e^{-4s}) / s^4$

2. A short dipole has a radiation resistance $R_r = \sqrt{(\mu_0 / \epsilon_0)} [(\lambda)^2 / 6\pi]$ ohms. Find A_{em} the maximum effective aperture of this dipole.

(a) $0.075 \lambda^2$

(b) $0.200 \lambda^2$

(c) $0.180 \lambda^2$

(d) $0.089 \lambda^2$

(e) $0.102 \lambda^2$

3. $y_{11} = 0.25$ mho, $y_{12} = -0.05$ mho, $y_{22} = 0.1$ mho. Find the z-parameters.

(a) $z_{11} = 11.1 \Omega, z_{12} = z_{21} = 4.44 \Omega, z_{22} = 22.2 \Omega$

(b) $z_{11} = 11.1 \Omega, z_{12} = z_{21} = 2.22 \Omega, z_{22} = 4.44 \Omega$

(c) $z_{11} = 2.22 \Omega, z_{12} = z_{21} = 4.44 \Omega, z_{22} = 11.1 \Omega$

(d) $z_{11} = 4.44 \Omega, z_{12} = z_{21} = 1.11 \Omega, z_{22} = 22.2 \Omega$

(e) $z_{11} = 4.44 \Omega, z_{12} = z_{21} = 2.22 \Omega, z_{22} = 22.2 \Omega$

4. What percent of the total volume of an iceberg floats above the water surface? Assume the density of ice to be $57.2 \text{ lb}_m/\text{ft}^3$, the density of water $62.4 \text{ lb}_m/\text{ft}^3$.

(a) 5 percent

(b) 40 percent

(c) 35 percent

(d) 10 percent

(e) 20 percent

5. Let $A = \begin{bmatrix} 22 \\ 12 \end{bmatrix}, B = \begin{bmatrix} 31 \\ 42 \end{bmatrix}$. Find AB .

(a) $AB = \begin{bmatrix} 76 \\ 75 \end{bmatrix}$

(b) $AB = \begin{bmatrix} 96 \\ 65 \end{bmatrix}$

(c) $AB = |76| |45| |138|$

(d) $AB = |96| |55| |138|$

(e) $AB = |86| |45| |138|$

True / False (10 marks)

Answer True or False

6. True or False: The Laplace transform of $f(t) = t^n$ is $n! / s^{n+1}$, which incorrectly omits a factor of s .
7. True or False: A phase shift of 30 in a phase-shift keying system ensures synchronization, resulting in a phase shift of 30.
8. True or False: The electrostatic energy of a point charge within a conducting shell is calculated using the formula $U = \frac{1}{2} \int \rho(r) \phi(r) dV$ (where ρ is the charge density and ϕ is the potential).
9. True or False: The one cycle surge current rating of an SCR with a half cycle surge current of 30A is 30A.
10. True or False: The curl of the vector field $\mathbf{a} = x^2y^2\mathbf{i} + 2xyz\mathbf{j} + z^2\mathbf{k}$ at the point (1, -2, 1) is (0, 4, 0).

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the number of isotopes of an element influences the determination of the number of rings in the Bohr model, and analyze its implications for engineering applications.
12. Evaluate the surface integrals $\int \int_S \mathbf{F} \cdot \hat{\mathbf{s}} ds$ and $\int \int_S \mathbf{F} \times \hat{\mathbf{s}} ds$ for the vector field $\mathbf{F} = kr^n \hat{\mathbf{e}}$ over a sphere of radius a centered at the origin. Discuss the implications of your results.

End of Paper